JCI Grand Baie

MauZanimo

Digital Pet Adoption Platform

Technical Documentation & Presentation Guide

JCI Grand Baie | Mauritius | Version 1.0.0 | April 2026

1. Project Overview

MauZanimo is a digital pet adoption platform built and powered by JCI Grand Baie. The app connects stray animals across Mauritius with loving families, volunteers, and animal welfare organizations through a clean and accessible mobile experience.

The platform was built as a social impact technology project to address the growing issue of stray animals in Mauritius. Rather than relying on word of mouth or informal social media posts, MauZanimo provides a structured, transparent, and community-driven solution.

1.1 Mission Statement

To leverage technology to reduce the number of stray animals in Mauritius by connecting pets with loving homes, supporting community volunteers, and building a data-driven animal welfare ecosystem.

1.2 Project Details

App Name	MauZanimo
Platform	Android Mobile Application (Flutter)
Backend	Firebase (Firestore, Authentication)
Languages Supported	English and French
Organisation	JCI Grand Baie, Mauritius
Version	1.0.0
Launch Target	Q2 2026
Distribution	Google Play Store

2. Technical Architecture

2.1 Technology Stack

Layer	Technology	Purpose
Frontend	**Flutter 3.41**	Cross-platform mobile app development
Database	**Firebase Firestore**	Real-time NoSQL database for pets, inquiries, events
Authentication	**Firebase Auth**	Email/password and Google Sign-in
State Management	**Provider**	Language preferences and app-wide state
Local Storage	**Shared Preferences**	Onboarding status and language selection
External Links	**URL Launcher**	WhatsApp integration and external links

2.2 Firebase Collections

- pets - All pet listings with profile details, health status and adoption status
- inquiries - Adoption requests submitted by users with applicant details
- users - Registered user profiles linked to Firebase Auth UIDs
- favourites - Saved pets per user for the Saved tab
- stories - Success stories shared by adopters
- lostfound - Lost and found animal reports posted by the community
- events - Upcoming adoption events managed by the admin
- volunteers - Volunteer applications submitted through the app

3. Feature Overview

3.1 User Features

Feature	Description
Language Selection	On first launch, users choose English or French. The preference persists across all sessions.
Onboarding	4-slide visual introduction shown only on first launch. Fully bilingual.
User Registration	Create an account with name, email, phone and password.
Login & Password Reset	Secure email/password login. Forgot password sends a reset email automatically via Firebase.
Pet Listings	Browse all available pets with photos, type filters and a live search bar.
Pet Detail	Full pet profile with health status, description, adoption button and WhatsApp contact.
Save / Favourites	Tap the heart icon to save any pet. Saved pets appear in the dedicated Saved tab.
Adoption Inquiry	Structured form for adoption applications. All inquiries stored in Firestore.
WhatsApp Contact	One-tap button on each pet profile to message the owner directly on WhatsApp.
List Your Pet	Any registered user can submit their pet for adoption via the floating button.
Lost & Found	Community board to report lost or found animals. Filterable by Lost or Found.
Adoption Events	Calendar of upcoming adoption drives managed by the admin.
Success Stories	Adopters share their story. Stories appear in a public feed to inspire others.
Volunteer	Users submit a volunteer application through the drawer menu.
Donate	Information on how to donate to support the platform and animal welfare.
Partners	Directory of veterinary clinics, shelters and NGO partners.
Profile	View profile info and track all adoption inquiry statuses.
Bilingual Interface	Full app switches between English and French via the hamburger menu toggle.

3.2 Admin Features

Feature	Description
Hidden Admin Access	Long-press the paw icon on the login screen to reveal the admin dashboard.
Dashboard Overview	Real-time stats: total pets, inquiries, adoptions and pending requests.
Add New Pet	Full pet registration form with photo URL, health status, type, gender and location.
Manage Inquiries	View all adoption applications and approve or reject them individually.
Manage Events	Create and manage adoption events directly via Firebase Firestore.

4. Application Screens

The following screenshots show the MauZanimo application running on a mobile device. All screens are from the live version of the app.

4.1 Splash & Language Selection

On first launch, users see the MauZanimo splash screen followed by a language selection screen featuring the JCI Grand Baie logo. Users choose English or French before proceeding to onboarding.

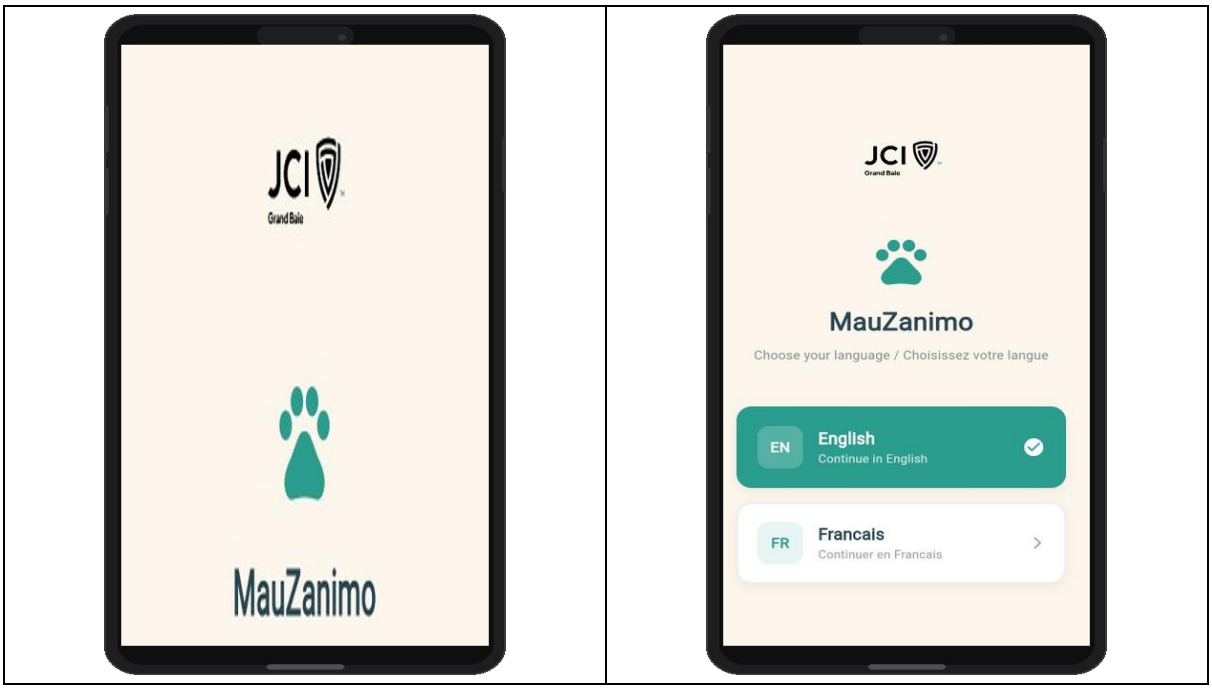

Figure 1 - Splash Screen (left) and Language Selection Screen (right)

4.2 Onboarding

A 4-slide carousel shown to new users only. Each slide uses a coloured icon, title and description to introduce MauZanimo's key features. Content is fully bilingual based on the user's language choice.

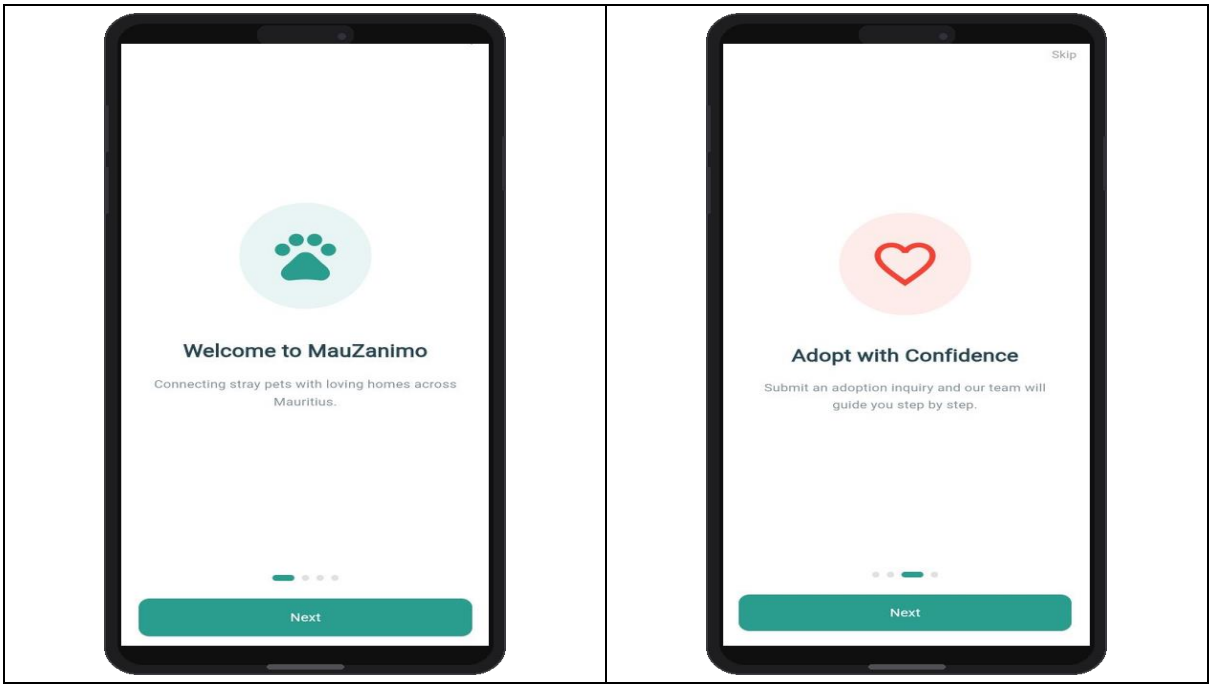

Figure 2 - Onboarding Slides: Welcome (left) and Adopt with Confidence (right)

4.3 Login Screen

Clean and minimal login with the MauZanimo paw logo, email and password fields, a Forgot Password link, and a Sign Up option for new users. Long-pressing the paw icon opens the hidden admin dashboard.

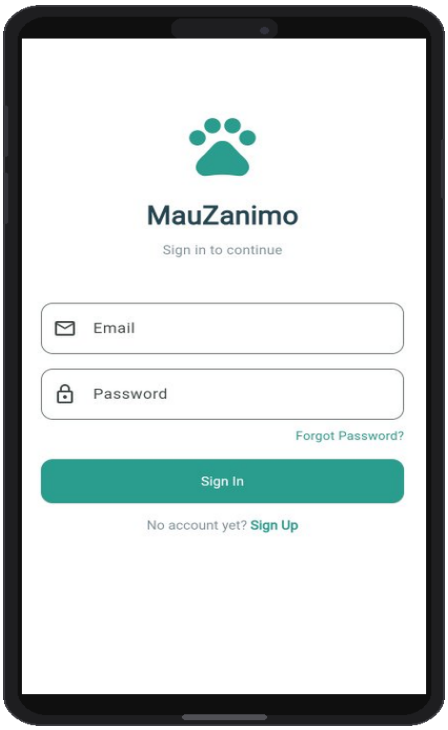

Figure 3 - Login Screen

4.4 Home Screen - Pet Listings

The main screen of the app. Features a teal header with search bar, filter chips (All, Dogs, Cats, Others), a real-time grid of available pets, and a floating List a Pet button for users to submit their own animals.

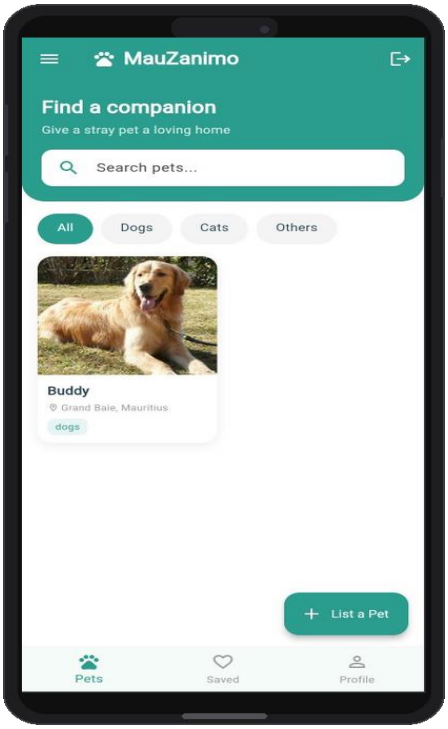

Figure 4 - Home Screen with Pet Listings

4.5 Pet Detail Screen

A full pet profile with expandable photo header, adoption status badge, location, type and age chips, health indicators (Vaccinated / Sterilized), an adoption inquiry button, and a WhatsApp contact button. The heart icon in the top right saves the pet to Favourites.

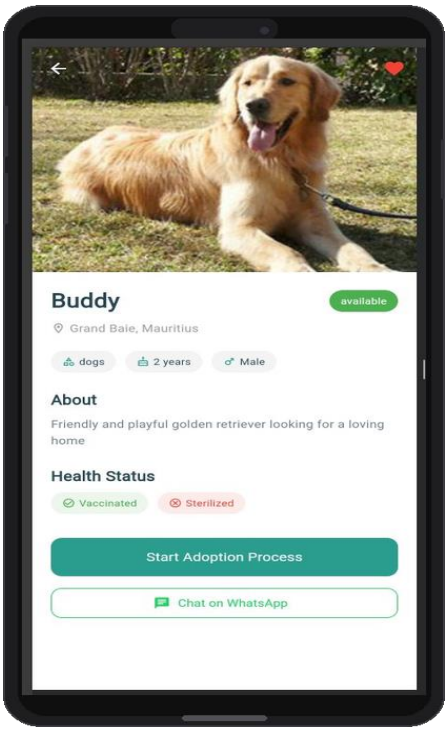

Figure 5 - Pet Detail Screen (Buddy the Golden Retriever)

4.6 Adoption Inquiry Form

A structured form where users fill in their full name, email, phone number, and a personal message. On submission the inquiry is stored in Firebase and immediately visible in the admin dashboard for review.

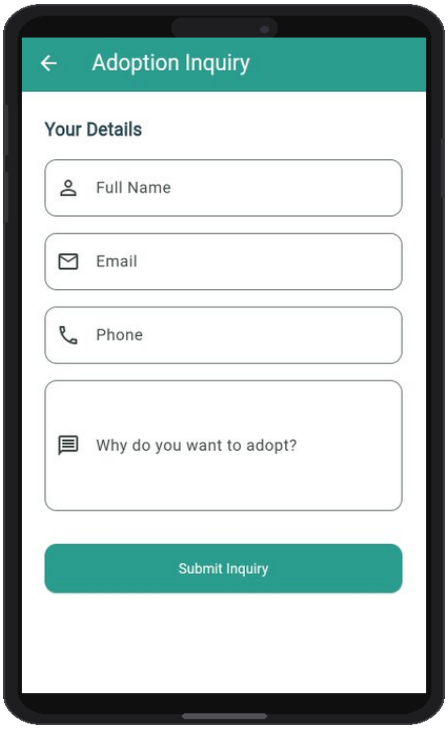

Figure 6 - Adoption Inquiry Form

4.7 Navigation Drawer

The hamburger menu is accessible from the top-left icon on any main screen. It shows the JCI Grand Baie logo, the logged-in user's email, and links to all major sections including Lost & Found, Adoption Events, Success Stories, Donate, Partners, Volunteer, About and Contact Support. A language toggle is available at the bottom.

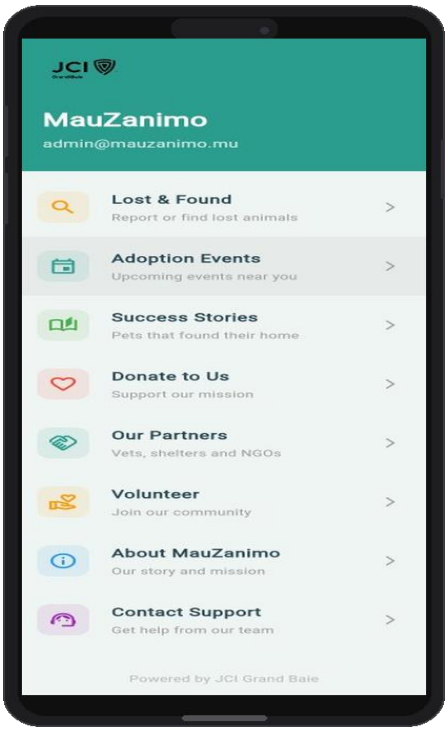

Figure 7 - Navigation Drawer (English and French toggle visible)

4.8 Lost & Found

A community board where users can report lost or found animals in their area. Posts are filterable by All, Lost, or Found. Each report includes a description, last seen location, animal type and contact number.

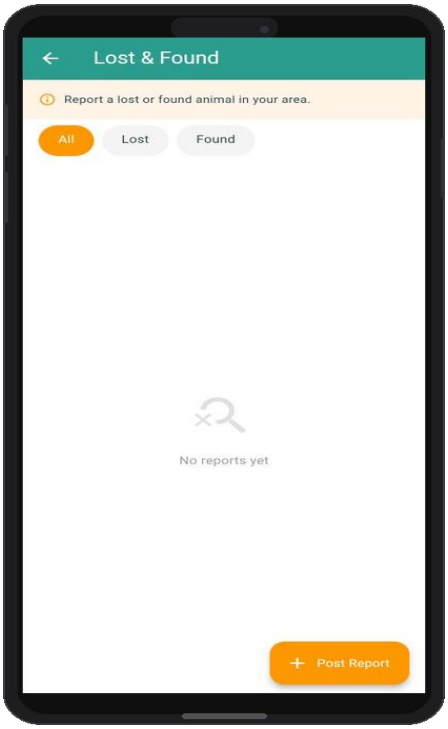

Figure 8 - Lost & Found Screen

4.9 Admin Dashboard

Accessible to admin users only via the hidden long-press gesture on the login screen. Displays four real-time stat cards (Total Pets, Inquiries, Adopted, Pending) and quick action buttons to add new pets or manage adoption inquiries.

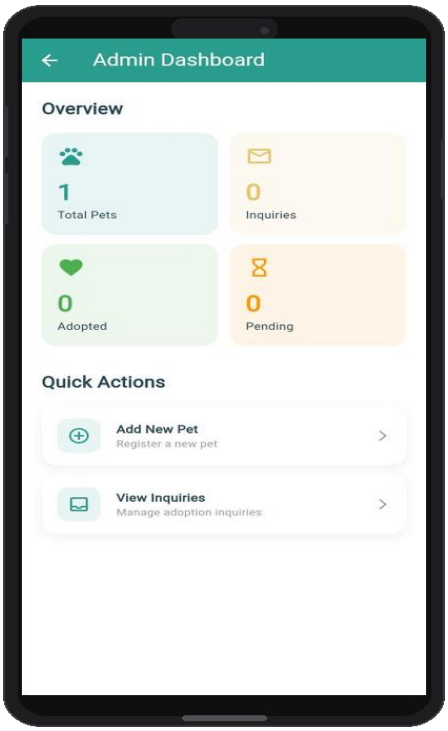

Figure 9 - Admin Dashboard

5. User Journey

The following outlines the complete journey a new user takes through MauZanimo from first download to successful adoption inquiry.

#	Step	What Happens
1	**Download & Launch**	User downloads the app from the Play Store and opens it for the first time.
2	**Language Selection**	User chooses English or French. This preference is saved for all future sessions.
3	**Onboarding**	User swipes through 4 slides explaining MauZanimo's mission and features.
4	**Create Account**	User registers with their name, email, phone and password.
5	**Browse Pets**	User browses the pet grid, uses filters or searches by name or location.
6	**View Pet Profile**	User taps a pet to see full details including health status and description.
7	**Save Pet**	User taps the heart icon to save the pet to their Saved tab for later.
8	**Submit Inquiry**	User taps Start Adoption Process and completes the adoption inquiry form.
9	**Admin Review**	Admin receives the inquiry, reviews it, and approves or rejects it.
10	**User Notified**	User checks their Profile tab to see the updated status of their inquiry.

6. Development Roadmap

6.1 Completed - Version 1.0.0

- Language selection screen (English and French)
- Animated splash screen with JCI Grand Baie branding
- 4-slide bilingual onboarding carousel
- User registration, login and password reset
- Real-time pet listings with search and type filters
- Full pet detail screen with health status indicators
- Save/Favourite pets per user
- Adoption inquiry submission and admin review workflow
- WhatsApp direct contact button on pet profiles
- User-submitted pet listings (List a Pet)
- Lost & Found community board with type filter
- Adoption Events section
- Success Stories feed
- Volunteer application form
- Donate and Partners information screens
- Bilingual navigation drawer with language toggle
- Hidden admin dashboard with stats and inquiry management

6.2 Planned - Version 1.1.0

- Push notifications for inquiry status updates
- In-app chat between adopters and pet owners
- Edit and delete pet listings for admin
- Mark pet as adopted directly from inquiry approval
- Analytics dashboard with adoption rate metrics
- Social sharing - share pet profiles to WhatsApp and Facebook
- Pet of the Week feature highlighted on the home screen

6.3 Future Vision - Version 2.0

- National scalability - onboard other animal welfare organisations across Mauritius
- Vet appointment booking through partner clinics
- Pet health tips and educational content section
- Adoption checklist for new pet owners
- Creole language support
- Regional filters by district (Grand Baie, Port Louis, Curepipe, etc.)
- Referral and community badge system

7. Deployment Guide

7.1 Play Store Submission Steps

1. Generate a release keystore using the keytool command
2. Configure signing in the Flutter build.gradle file
3. Run: flutter build appbundle --release to generate the .aab file
4. Create a Google Play Console account (one-time $25 USD fee)
5. Upload the .aab file and complete the store listing (description, screenshots, category)
6. Fill in the content rating questionnaire
7. Submit for Google review (typically 3-7 business days)

7.2 Required Assets for Submission

- App icon: 1024 x 1024 px PNG
- Feature graphic: 1024 x 500 px
- At least 2 phone screenshots
- Short description (80 characters max)
- Full description (4000 characters max)
- Privacy policy URL (required by Google)

7.3 Critical Files to Back Up

The following files must be stored securely. Without these, the app cannot be updated on the Play Store.

- mauzanimo.keystore - The signing key for Play Store releases

- key.properties - The keystore credentials file
- firebase_options.dart - Firebase project configuration
- Full source code - Push to GitHub for safekeeping

8. Project Contact

Organisation	JCI Grand Baie
Project	MauZanimo - Digital Pet Adoption Platform
Location	Avenue des Goyaviers, Quatre Bornes, Mauritius
App Email	jcigrandbaie@gmail.mu